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Student Number: 201538542

Module Title and Number: LAW3035 Long Dissertation (Criminal Justice and Criminology)

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Date of Meetings	Record of Meeting
07/11/2023	Discussion of 500 word plan submitted prior to the meeting with a particular focus on the methodology of the proposed research, including the strategy for recruitment, the use of an online survey compared to the conduction of a lab in person, and the data collection process.
06/02/2024	- Overview of recruitment poster, information sheet, and online survey with suggestions on what to modify or improve; - Discussion of previously conducted pilot study and of any potential ethical issues involved in the study; - Request of fulfilment of ethics form and data management plan.
05/03/2024	- Feedback regarding literature review; - Discussion of data analysis plan to begin analysis post-meeting.
21/03/2024	- Feedback on results; - Discussing key points to be made in the discussion section.

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Student Identification Number:

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Module Code:	LAW	3	0	3	5
Module Title:	Long Dissertation (Criminal Justice and Criminology)				
Declared Word Count:	11814				

Looking beyond race: the role of contact in reducing the Cross-Race Effect among White individuals in England and Wales



*Dissertation submitted for the fulfilment of the requirements of
BSC (Hons) Criminal Justice and Criminology (Quantitative)*

April 2024

School of Law
University of Leeds

Declaration

	Please Tick
I declare that this dissertation has been composed solely by myself and that it has not been submitted, in whole or in part, in any previous application for a degree. Except where states otherwise by reference or acknowledgment, the work presented is entirely my own.	✓

Date: 22/04/2024

Acknowledgments

First and foremost, I would like to thank the participants of this research study who took their time to fill out my survey, without whose contribution, there would have been no way to obtain the data necessary to undertake the research. I would also like to thank my family and friends who took part in helping me share my survey with the public, aiding the process of recruitment.

A special thank you to my dissertation supervisor, Toby Davies, for his support and guidance throughout the entire process, from start to finish.

Finally, I would like to acknowledge the support given to me by my parents, who constantly kept me grounded and reminded me of my worth in extremely stressful times. A special thank you also to my friend Bryanna Heaton, for her subconscious support during the long hours spent working together, and for always being present when in need of a rant.

Abstract

The Cross-Race Effect (CRE) explains the superior capability of individuals to identify faces of their own race, compared to faces of other races. Among the many theories which have been made in an attempt to explain this phenomenon, the contact hypothesis stands out for its high reliability, explaining that this recognition deficit is due to a limited amount of interracial contact. The CRE has a major impact on cross-race eyewitness identification, potentially leading to wrongful convictions, particularly of ethnic minorities. While much research has already confirmed the consistency of the CRE, questions remain surrounding its causes. The consistency of the contact hypothesis, which could explain one of these causes, remains to be confirmed. This research aims to establish the presence of the CRE and test the validity of the contact hypothesis among White individuals living in England and Wales whose levels of contact with other racial groups were assessed using ethnic group population data. To obtain the data, an online recognition task was devised. The study finds that, while the consistency of the CRE remains proven, the contact hypothesis is not valid when the type of contact experienced is that supposed in a multiracial society. The findings suggest further research needs to be undertaken to provide an improved understanding of the type of contact which needs to be experienced in order for the contact hypothesis to improve recognition abilities of individuals for other-race faces.

Contents

Acknowledgments.....	5
Abstract.....	6
CHAPTER 1: Introduction	9
CHAPTER 2: Literature Review.....	12
Introduction	12
The Cross-race Effect within the Criminal Justice System	12
A lack of evidence.....	12
The importance of experience	14
The contact hypothesis.....	16
England and Wales as multiracial societies	17
Artificial Intelligence in research	17
Conclusion	17
CHAPTER 3: Methodology	19
Introduction	19
Methodological approach	19
Sampling.....	19
Data collection.....	20
Ethics.....	22
Data analysis.....	23
CHAPTER 4: Results.....	24
Recognition accuracy	24
Hits	24
False alarms	25
Level of contact.....	25
Recognition accuracy across race of stimuli	30
CHAPTER 5: Discussion.....	35
The CRE in England and Wales.....	35
The contact hypothesis, an effective explanation?	36
Limitations and recommendations for future research	39
CHAPTER 6: Conclusion.....	41
References	44
Appendices.....	48
Appendix A: Additional results.....	48
Appendix B: Ethics Form	48

Appendix C: Participant Information Sheet	58
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CHAPTER 1: Introduction

In 1984 in North Carolina, 22-year-old Jennifer Thompson-Cannino's apartment was broken into by a male who put a knife to her throat and raped her (Thompson-Cannino, 2000). Following Jennifer's report of the incident to the police, Ronald Junior Cotton, a man of African American heritage, was found guilty and sentenced to life in prison plus fifty-four years (Innocence Project, n.d.). The only evidence supporting Cotton's conviction was a flashlight found in his apartment, similar to the one held by the attacker at the scene, and the rubber from one of his shoes, consistent with that which had been found in Jennifer's apartment following the attack (Innocence Project, n.d.). His conviction rested mostly on the eyewitness testimony provided by Jennifer who stated that:

'I studied every single detail on the rapist's face. I looked at his hairline; I looked for scars, for tattoos, for anything that would help me identify him' (Thompson-Cannino, 2000).

It was only 11 years later that DNA was collected and tests were ran using evidence gathered from the rape scene (Thompson-Cannino, 2000). The results of the test found Ronald Junior Cotton innocent and another man, Bobby Poole, guilty (Thompson-Cannino, 2000). On the 30th of June 1995, Cotton was cleared of all charges and released from prison (Innocence Project, n.d.).

Robert Cotton's case is one of the many examples of wrongful convictions caused by cross-racial eyewitness accounts. The Innocence Project, an organisation created with the aim of freeing the innocent, and with the wider scope of creating a fair criminal justice system (Innocence Project, n.d.), has found that since 1992, almost 60% of people who received its help are Black (Innocence Project, n.d.). Some of the data provided by Gross, et al. (2022) in their report conducted in the United States, reveals that Black people are seven times more likely to be convicted of murder and eight times more likely to be convicted of sexual assault compared to White people. The report also explains that the largest cause of this racial disparity is due to the high likelihood of Black individuals being mistakenly identified by White individuals, leading to their wrongful convictions (Gross, et al., 2022). Indeed, while eyewitness testimony may seem like an extremely reliable source, many do not realise that face recognition memory is affected by a number of factors, such as viewpoint, (Hill, et al., 1997; O'Toole, et al., 1998; Favelle, et al., 2017) distinctiveness (Valentine, 1991; Chiroro & Valentine, 1995; Meissner, et al., 2005), and lighting (Johnston, et al., 1992), which could result in wrongfully accusing someone of a crime they did not commit. In particular, identification is profoundly affected by race, leading eyewitness testimony to be extremely unreliable in cases which involve cross-racial identification (Innocence Project, n.d.). Above all, White individuals are known to have exceptional difficulties identifying other subjects when they are not white (Innocence Project, n.d.), causing ethnic minorities to be the group which is most affected by mistaken identification in

eyewitness testimony. This increased capability individuals have of recognising faces of their own race compared to faces of other, less familiar races (Meissner & Brigham, 2001), is known as the Cross-Race Effect (CRE) or own-race bias (ORB) and is likely to be the cause of 42% of wrongful convictions caused by misidentification (Montana Innocence Project, 2024).

The major consequences the CRE can have during eyewitness testimony make this phenomenon an extremely important one within the context of the criminal justice system. Understanding the causes behind it, could lead to the adoption of preventative measures aimed at minimising its impact. This could reduce numbers of wrongfully convicted individuals, particularly those from ethnic minorities, who, like Ronald Junior Cotton, are forced to waste years of their life in prison for a crime they did not commit.

Much research in this area has already been conducted, consistently proving the CRE across several racial groups and cultures. Research on the CRE has confirmed the reliability of the phenomenon across four levels: studies, racial and ethnic groups, memory tasks, and individuals and testing occasions (Meissner & Brigham, 2001). However, research attempting to understand its causes is a lot more conflicting in its findings (Meissner & Brigham, 2001).

The contact hypothesis is one of the many potential causes of the CRE which have been identified in research and is the theory upon which most research has agreed on (Slone, et al., 2000). This hypothesis presupposes that the difficulty which presents itself during the recognition process of other-race faces is caused by limited contact and exposure with and to individuals of other races. Indeed, according to this theory, increased interracial contact can have an impact on the way faces are encoded and memorised, leading to differences in recognition abilities (Hancock & Rhodes, 2008). Studies testing the contact hypothesis are especially contrasting in their findings (Chiroro & Valentine, 1995), stemming mostly from the difficulty in understanding the type of contact which can be successful in improving recognition of other-race faces, and the type which is not as effective (Ng & Lindsay, 1994). The inconsistency of these findings has resulted in an unclear understanding of this hypothesis and the way interracial contact can influence recognition.

In order to address the issues mentioned above, this study contributes to the wide body of existing research studying the CRE and the contact hypothesis by answering the following research question:

‘To what extent can the contact hypothesis provide an explanation for the Cross-Race Effect in White individuals residing in England and Wales?’

In particular, two hypotheses are made:

- That the CRE exists across White individuals living in England and Wales;

- That the contact hypothesis is a valid explanation for its occurrence.

As previously mentioned, the CRE has been referred to in different ways throughout research, including 'cross-race effect', 'other-race effect', and 'own-race bias'. For consistency purposes, the phenomenon will be referred to as 'cross-race effect' in its abbreviated form CRE throughout this whole paper.

This dissertation is divided into a total of six chapters, including the current introduction. The following chapter consists of a literature review which will critically analyse and discuss existing literature surrounding the topics of the CRE and the contact hypothesis. Chapter 3 outlines the methodology used throughout the conduct of this study, and this is followed by a chapter providing the results of the statistical analysis conducted on the data obtained from the study. The final two chapters consist of a discussion interpreting the results previously discussed in chapter 4, and a conclusion summarizing the main findings of the research, in addition to acknowledging limitations of the study and possibilities for future research.

CHAPTER 2: Literature Review

Introduction

This section will provide the reader with an overview of the existing literature discussing the CRE within the context of the criminal justice system, and the contact hypothesis. Following a funnel type layout, this section will first consider the CRE and how it relates to the fields of criminal justice and criminology. Next, the numerous hypotheses attempting to explain this phenomenon will be analysed with a particular focus on the contact hypothesis. Finally, the relevance of the literature in relation to this particular research study will be clarified.

The Cross-race Effect within the Criminal Justice System

For a significant period, the likely inaccuracy of eyewitness testimony in court and the probability of it leading to misidentification, have been recognised (Read, 2006; Wilson, et al., 2013). In his research paper, Read (2006) synthesises existing knowledge in regards to eyewitness testimony and its unreliable link to identification, but makes a significant contribution to this topic by focusing on estimator variables (Wells, 1978) in eyewitness misidentification. Indeed, most research in this area tends to focus on system variables, variables which can be controlled by the legal system (Wells, 1978). On the other hand, estimator variables include all those factors which, in a court setting, are not under the control of the legal system and could lead to eyewitness error. These could be 'the specific capabilities of human observers as a function of their genetics, age, and backgrounds, the lighting at the scene of the crime, and so forth' (Read, 2006, p. 525). Among these factors is the CRE. Wilson et al. (2013) emphasise the importance of developing an in-depth understanding of the CRE in order to be able to successfully eliminate the influence of this phenomenon in court settings. According to their theoretical research, even though laboratory studies are limited to a very different scenario compared to that of a court setting, these can still be applied to the wider context of the CRE and its impact on eyewitness misidentification because of their capacity to test the CRE across different cultures and ethnicities (Wilson, et al., 2013).

A lack of evidence

A useful way of understanding the numerous theoretical approaches which have been adopted when studying the CRE, is that presented by Meissner and Brigham (2001) in their meta-analytic review. This approach contains a number of theories explaining the CRE and presents these in a comprehensive manner making this a fitting foundation for the present study. The two authors categorise explanations for the CRE into four groups: 'racial attitudes', 'physiognomic homogeneity', 'interracial contact' and 'perceptual learning' (Meissner & Brigham, 2001, pp.7-9). The first explanation assumes that those with less prejudiced attitudes towards other racial groups are less likely to experience difficulties in differentiating between other-race faces because they are

motivated to correctly identify these (Meissner & Brigham, 2001). The authors of this review state that racial attitudes have not been found to have a direct link with the CRE. However, it is worth noting that motivation can play a role in face recognition, whether it stems from racial attitudes or not. This was demonstrated by Hugenberg, et al. (2007). In this quantitative study, the authors attempt to explain the CRE through social cognitive theories, specifically the feature-selection model presented by Levin (2000). This model supposes that difficulties in recognition arise because of certain specific features which individuals select when encoding information about faces (Levin, 2000). Put into other words, the model assumes that the CRE is caused by 'differences in social cognitions' (Hugenberg, et al., 2007, p.335) which occur when differentiating between members of ones' own race and those of another race, causing the latter to be categorised. In the context of the CRE, categorisation refers to the classification of faces based on their belonging to a certain social group rather than because of individual characteristics which define them, a process known as individuation. Based on the model provided by Levin (2000), Hugenberg, et al. (2007) hypothesise, and are able to prove, that motivating individuals to use individuation when identifying out-group members can make them more aware of their tendency to categorise and can diminish the CRE. However, one must note that the study conducted by Hugenberg, et al. (2007) does present limitations: these include the limited generalisability of the findings caused by participants mostly being European American individuals, and the potential bias stemming from informing participants about the CRE, leading them to potentially behave in ways they think are expected from them. Nonetheless, the findings of this study do highlight the potential importance of motivation in relation to the CRE and there are other studies which also support this. An example is that of Wilson et al. (2013), who emphasise the importance of motivation in successfully individuating people. Importantly, the potential of motivation in reducing the CRE is only expected when it is paired with experience (Wilson, et al., 2013), a factor the importance of which will be made clear later in this review.

The second explanation presented by Meissner and Brigham (2001), that of 'physiognomic homogeneity' (p.7) suggests that individuals of other racial groups may show less variability in comparison to those of ones' own race. Once again, very little evidence has been found in support of this (Meissner & Brigham, 2001; Wong, et al., 2000; Sporer, 2001). Because of the lack of evidence in support of the explanations of 'racial attitudes' and 'physiognomic homogeneity' (Meissner & Brigham, 2001, p.7), these explanations are set aside in order to focus on the 'interracial contact' and 'perceptual learning' (Meissner & Brigham, 2001, pp. 7-8) explanations, deemed to be more important, particularly in the context of this study.

The importance of experience

A number of studies have tested the influence of interracial contact on the CRE and have found a link between the two (Meissner & Brigham, 2001). The idea underpinning these studies is that the quantity and quality of interracial contact an individual is subjected to can influence his/her ability to differentiate between members of different racial and ethnic groups (Meissner & Brigham, 2001). The perceptual learning hypothesis, also known as perceptual expertise hypothesis, addresses this idea by attempting to understand exactly how interracial contact can influence the CRE, suggesting that individuals use specific cues to recognise faces of their own race which may not be applicable to the recognition of faces of other races (Meissner & Brigham, 2001). It is however possible for these cues to be improved through experience and contact to the point of being applicable to other-race faces (Meissner & Brigham, 2001). MacLin et al. (2004) sought to test the perceptual expertise hypothesis in their quantitative study using participants who had knowledge in the area of basketball (classified as 'experts') and participants who did not (classified as 'novices') (MacLin, et al., 2004). While in the first experiment participants were simply shown faces of lay people, experiment two used photos of famous African American basketball players and famous African Americans in the recognition task. In addition, the stimuli were presented both in an upright and in an inverted manner (MacLin, et al., 2004). The findings of MacLin et al. (2004) lend partial support to the perceptual expertise hypothesis by demonstrating a higher inversion effect for same-race faces, meaning participants had more difficulty recognising these when they were presented in an inverted manner rather than in an upright position (MacLin, et al., 2004). In experiment two, 'novices' were more affected by the inversion effect for same-race faces than 'experts' were, suggesting expertise in recognising individuals from particular groups is paired with a different cognitive processing of faces. Indeed, the 'experts' groups did not show an inversion effect for other-race faces, suggesting a potential different processing method compared to the 'novices' group (MacLin, et al., 2004). What is thought to be particularly interesting about MacLin et al.'s (2004) study is the suggestion that the perceptual expertise hypothesis may be satisfied simply through ones' contact with the stimuli through the media, although this remains to be proven further.

The study by MacLin et al. (2004) is also a good way of demonstrating how the numerous theories attempting to explain the CRE feed into each other. Indeed, the findings which have been discussed highlight the influence of different cognitive processes on face recognition. The theoretical assumption explaining these is known as the configural featural hypothesis and generally assumes that individuals tend to rely on configural features (i.e. how features relate to one another and come together as a whole) when recognising faces of their own race, but focus on single features when identifying other-race faces (Meissner & Brigham, 2001). Michel et al. (2006) identify the processing

of same race faces as holistic processing and find support for this hypothesis in their quantitative cross-cultural study involving Caucasian and Asian participants with increased experience with same-race individuals. While Michel et al. (2006) explain that the reason in choosing such participants is in order to successfully and equally compare the performance of both ethnic groups, this choice may have come with certain limitations, such as the potential for cultural biases. In addition, including individuals with increased other-race experience in the study could have added to the findings and conclusions of the research. Nonetheless, the use of a face-composite paradigm to study the CRE was an innovative aspect of the study. This technique involves aligning the top half of one face with the bottom half of another face, causing individuals to have difficulty recognising the top half. This is the case particularly when the face is of the same race as the observer (Michel, et al., 2006). The authors of this paper align the difficulty of recognising same-race faces caused by the face-composite paradigm with the tendency to perceive these faces more holistically (Michel, et al., 2006). This finding is further supported in Hahn et al.'s (2011) experiment which aimed to test the configural-features hypothesis by using Thatcherization, a technique consisting of the rotation of the eyes and mouth by 180°. This study is particularly valuable because of the innovative application of this technique involving the use of electrophysiology and event-related potentials (ERPs) to demonstrate the validity of the configural expertise theory. Specifically, the study considers three ERPs (P1, N170, and P250) all associated with face processing and one (P250) associated with familiarity in face processing. Using measures of grotesqueness as an indicator of the disruption caused by the rotation of the facial features, Hahn et al. (2011) found that Thatcherization had a larger impact on the processing of same-race faces.

Finally, Valentine (1991) proposes a unique method of comprehending the process of encoding faces and their consequential recognition. As explained by Valentine (1991), 'individual faces are assumed to be encoded as a point in a multidimensional space, defined by dimensions that serve to discriminate faces' (p.161). Based on this space, two models are created: a norm based coding model and an exemplar based norm model. The first model predicts that 'faces are encoded in terms of their deviation from a single general face norm or prototype located at the origin of the space' (Valentine, 1991, p. 167), and that therefore other-race faces are harder to recognise because they are further from the norm. In the exemplar based norm model, there is no standard norm and faces are compared directly to faces which have already been stored in the space (Valentine, 1991). While the differences between the two models are not major and neither can be discriminated against as they both present valid ideas, the exemplar based norm model has greater potential to explain the importance of the link between race and experience in facial recognition. Indeed, this model suggests that increased experience with other-race faces can improve the classification of these

because of the increased density of these types of faces in the multidimensional space (Valentine, 1991). This goes in line with what the contact hypothesis explains, which is that limited contact with individuals from different racial groups can have a negative effect on our capability of recognising these individuals (Meissner & Brigham, 2001).

The contact hypothesis

The contact hypothesis is a longstanding, rigid one, which research has investigated profusely; however, consistent results which successfully support this theory are yet to exist (Chiroro & Valentine, 1995). The study conducted by Chiroro and Valentine (1995) supports the suitability of the exemplar-based norm model to explain the contact hypothesis. In their quantitative study, following a cross cultural design, involving the use of African and White subjects, Chiroro and Valentine (1995) sought not only to investigate the contact hypothesis, but also to understand the way distinctiveness interacts with both the race of the face and that of the individual. The authors were able to successfully prove an interaction between race of subject, contact group, race of face and distinctiveness, demonstrating that distinctive faces are more easily recognisable than typical faces (Chiroro & Valentine, 1995). This supports Valentine's (1991) prediction regarding distinctiveness, explaining that in the exemplar based norm model, distinctive faces are encoded further away from the norm, making the likelihood of them being misidentified lower.

A second finding made by Chiroro and Valentine (1995) was the way Black individuals were more affected by the contact hypothesis than White individuals were. This is something which, to the knowledge of this researcher, has not been suggested by any other studies. Indeed, in the literature discussed, the contact hypothesis has always been presented as having an equal effect on all groups of subjects participating in the studies.

As stated, findings around this theory are inconsistent, and there has been research which has found no significant interaction between the amount of interracial contact and recognition (Ng & Lindsay, 1994; Slone, et al., 2000). It is important to note however, that both studies which reported these results used a self-report questionnaire to assess participants' level of contact and experience with other racial groups. This type of method is believed to be a significant limitation to the studies because of the likelihood it has of being influenced by social desirability bias stemming from people over-estimating their level of contact with individuals of races different from their own (Walker & Hewstone, 2006). Because of the criticisms around the use of self-report questionnaires to assess levels of contact, this type of method was excluded from the present research project.

Research investigating the contact hypothesis has also found that quality of interracial contact is more important than quantity as a predictor of recognition ability (Sporer, 2001; Michel, et al., 2006;

Brigham & Malpass, 1985). However, when considering this implication, it can be complicated to assess what classifies as high quality contact and studies can be very subjective in the way they choose to interpret this. For instance, as stated previously, MacLin et al. (2004) found that contact with faces through the media can increase experience. But can the media truly be considered 'high quality contact'? This remains up for debate. As suggested by Sporer (2001), in order to successfully conclude if quality plays a more important role than quantity does, it would be necessary to find an objective measure to assess quality.

England and Wales as multiracial societies

The CENSUS data published in 2022 (Office for National Statistics, 2022) demonstrates the multiracial composition of England and Wales, with a total of 19 ethnic groups living in the countries. With such high numbers of individuals who identify as ethnic minorities, reducing the impact the CRE has on wrongful convictions of these groups is vital. For this reason, this research study aims to contribute to a deeper comprehension of the CRE by testing the validity of the contact hypothesis on a larger scale, namely in the whole area of England and Wales. To the knowledge of this researcher, although existing research has tested the CRE in England (Wright, et al., 2001; Wright, et al., 2003), it has yet to test this phenomenon and the contact hypothesis under the conditions adopted in this research study. Establishing whether its validity holds over such a wide area could lead to further support of this hypothesis, potentially making it a useful explanation of the CRE.

Artificial Intelligence in research

Since the release of ChatGPT in 2022, Generative Artificial Intelligence (GAI) and generative learning has become increasingly popular in academic research. According to Beyan & Rossy (2023), the generation of images through the use of text inputs in particular, has become more and more popular. Despite the increasing adoption of AI (Artificial Intelligence) tools in numerous fields, including architecture (Beyan & Rossy, 2023), medicine, education, media, entertainment, and academic research (Bengesi, et al., 2023), no study testing the CRE up to this date has used AI image generators to create faces for the recognition tasks typical of these studies. This research study takes this opportunity by employing BingAI (Microsoft Bing, n.d.) to create the stimuli used in the recognition task. It is believed this method could create stimuli equivalent in all conditions across the three racial groups, a crucial factor found to potentially impact the validity of results in studies testing the CRE (Banks, 1970).

Conclusion

In conclusion, it has been made clear that the CRE is a well proven phenomenon but that numerous difficulties remain in attempting to explain the specific reasons behind its occurrence. This area of research is one which is vital to the disciplines of criminal justice and criminology, as the CRE has

been demonstrated to be one of the leading causes of misidentification leading to wrongful convictions, particularly of ethnic minorities. Not having a full understanding of the reasons behind the CRE prevents the introduction of successful policy interventions that could potentially minimize numbers of wrongful convictions.

The following chapter will address the methodology used in this research study. This will include a discussion regarding the approach, the methods and the sampling techniques which were employed, how the data was collected and analysed, and a consideration of any ethical implications involved in this project.

CHAPTER 3: Methodology

Introduction

This chapter will discuss the methods used throughout the research process in order to test the existence of the CRE in England and Wales and its consistency with the contact hypothesis. More specifically, this section will describe and justify the methods employed before evaluating their suitability. This will include the sampling process and techniques, the use of an online survey to collect data, and data analysis. In addition, any potential strengths, and limitations of these will be reflected upon. Finally, any ethical issues which had the potential of arising during the research process, and how these were addressed, will be discussed.

Methodological approach

In order to answer the research question leading this project, a quantitative, deductive approach was utilised. A deductive approach is one in which theory guides research rather than the opposite (Clark, et al., 2021). Considering this project aims to test the validity of one of the many existing theoretical assumptions attempting to explain the social phenomenon of the CRE (i.e. the contact hypothesis), the use of a deductive approach was necessary. In addition, the positivist, objective approach that the quantitative method offers was viewed as being the best strategy to answer the research question. Indeed, the experimental design typically advocated by the positivist paradigm was believed to be the best method to test the existing assumptions regarding the CRE. This type of approach would have allowed a separation between the researcher and the participants, minimizing bias, and allowing for the production of objective data (Park, et al., 2020), something which could not have been achieved through the use of an interpretivist, constructionist approach, which places its emphasis on the importance of subjective experience (Chen, et al., 2011).

Sampling

To collect the data necessary for the purpose of this study, an online survey was devised and a sample of 70 White individuals aged between 18 and 64, residing in England and Wales, was recruited. The aim of this survey was to test White individuals' capability of recognising previously seen faces belonging to three different ethnic groups: White, Black Caribbean and Indian. The choice of using stimuli belonging to these particular groups was supported by background research regarding the population of England and Wales. Keeping in mind that the aim of this research study was to test the contact hypothesis, this type of data was necessary in the selection process involving the choice of ethnic groups to include in the recognition task, to be able to assume which groups White individuals would show greater capability of recognising. More specifically, White stimuli were selected because of the predominance of White individuals in England and Wales. Indeed, the latest data released by CENSUS (Office for National Statistics, 2022), reports that 81.7% of the population

in England and Wales is white. It was therefore assumed that White individuals would be more capable of recognising White stimuli. On the other hand, the other two ethnic groups, Black Caribbean and Indian, were selected for the opposite reasons. In fact, both these groups lack representation in England and Wales, with Black Caribbean individuals accounting for 1.0% of the population and Indians being slightly more common, representing 3.1% of the population (Office for National Statistics, 2022). Taking this into consideration, it was expected that White individuals would find it more difficult to recognise these stimuli, Black Caribbean stimuli in particular, because of the expected lack of contact with the two groups.

In order to select participants for the research study, three techniques of non-probability sampling were adopted: convenience sampling, snowball sampling, and criterion sampling. The first two sampling methods were used because of their accessibility to the researcher. Criterion sampling on the other hand, was needed to be able to obtain a sample fitting the ethnicity and residential requirements vital for the successful conduct of the study. The survey and the requirements to take part in the experiment were shared online via a number of social media platforms including Instagram, Facebook, Twitter, Snapchat, and WhatsApp. In addition, the website SurveyCircle (SurveyCircle, 2024) was employed to be able to reach a higher number of individuals. A poster was also created with details regarding the survey and shared around the campus of the University of Leeds, where students who were handed the poster were asked to further share the information.

It was recognised that having used three types of non-probability sampling techniques the sample had the potential of being subjected to bias, due to some individuals being more likely to be chosen than others (Clark, et al., 2021). For instance, the use of social media to share the survey could have enhanced the chances of a younger audience taking part. This can be explained through the wider use of these platforms by this type of audience. Because of this, it must be taken into account that the results of this research study are not generalisable to the wider population. Adding to the limitations related to generalisability was the representativeness of the sample which, having derived from three non-probability sampling methods, made knowing which population the sample was representative of impossible (Clark, et al., 2021).

Data collection

As previously stated, the data collection tool used was that of an online survey. While initially it was thought to conduct this research in a laboratory setting, which is what most studies regarding the CRE resort to (Wright, et al., 2001), this was decided against. While a laboratory setting would have enabled the monitoring of participants and the certainty that these were correctly responding to the task given to them, this setting would have required participants to be present in person, which was

thought to potentially limit the number of individuals willing to take part in the study for reasons related to practicality. On the contrary, conducting the study online allowed access to a wider and more diverse population who was assumed would be more prompted to complete the experiment if it was available to access from an environment of one's choice. Additionally, being that most studies on the CRE have already been conducted in a laboratory setting (Wright, et al., 2001), the use of an online survey was considered to be a valuable addition to this area of research.

Prior to the creation of the survey, 48 photos of individuals were created using the Image Creator tool offered by BingAI (Microsoft Bing, n.d.). Of the photos created, 16 were requested to be of White individuals, 16 of Black Caribbean individuals, and 16 of Indian individuals. To create these, the following prompt was used: 'A portrait photo of a *one of the three ethnic groups* individual wearing a black t-shirt and looking straight ahead'. The choice of an AI tool to create these images was supported by a number of advantages. Firstly, the avoidance of any ethical issues related to privacy stemming from the use of using pictures of real-life individuals who would have been subjected to the possibility of being recognised. Secondly, considering the contemporaneity of this tool, this choice of method was thought to be a potential interesting new addition in this area of research. Thirdly, the use of such a tool provided cost and time efficiency, allowing for a quick, precise, and effective way of obtaining the stimuli needed for the study. Finally, for the statistical measure of D-prime used in this research (of which information is provided further on) to be accurate, and for distractors to be compared, these need to be equivalent in all conditions (Banks, 1970). Using an AI tool to generate the stimuli and inserting the same prompt each time, was believed to be the best method to employ in order for this condition to be satisfied.

A potential limitation of the use of an AI tool was also acknowledged. As mentioned previously, the prompt inserted requested the creation of faces of a particular ethnic group. However, there was no guarantee the images generated were genuinely of these groups. Therefore, at the beginning of the survey, one question was inserted for each ethnic group asking participants to code the ethnicity of the stimuli being shown to them. In addition, a pilot study was conducted prior to the release of the survey in order to test not only the perceived ethnic groups of the stimuli by participants, but also to ensure the whole experiment worked efficiently and was clear in its instructions.

The online survey was divided into two parts, each created with a different software. The first part was designed with the use of PowerPoint and simply included a total of 24 stimuli: 8 White, 8 Black Caribbean, and 8 Indian. The faces shown in the presentation were the same for each participant. The second part of the survey was created with Microsoft Forms and included a total of 48 photos of individuals: 24 of these were the same as those in the PowerPoint presentation (defined as target

photos), while 24 of them were new (defined as distractors). Participants were asked to watch the PowerPoint presentation before moving on to the survey. No instructions were given besides highlighting that all that was requested during the viewing of the presentation was to observe the faces shown. The presentation allowed participants to view each photo for 5s (Sporer, 2001) before moving on automatically. Once the presentation had been viewed, participants were required to complete the survey asking them whether they recognised the person in the photo being shown to them as one of the previously seen faces in the PowerPoint presentation. Participants were required to answer with 'Yes' or 'No' and in this case the photos were not timed. Additionally, the survey collected information regarding participants' ages. This characteristic was thought to potentially be a useful control variable later during the stage of analysis, as previous research has examined the CRE with reference to age and found variance in the ways individuals of different ages process faces (Brigham & Williamson, 1979; Pezdek, et al., 2003; Sangrigoli & De Schonen, 2004).

Due to the awareness of the likelihood of social desirability bias arising from the use of a self-report questionnaire (Walker & Hewstone, 2006) to measure levels of contact, no method beyond the use of the CENSUS data (Office for National Statistics, 2022) was employed.

As can be the case with any type of field research, the process described was not a smooth one. In fact, during the initial stages of data collection, obstacles were met. These mainly consisted in attracting an audience to complete the survey, as there was no previous experience from which to obtain insight. Fortunately, this situation was easily solved by searching for advice on recruitment and encouraging friends and family to share the survey with as many people as possible. This led to a snowball process of attracting more and more participants.

Ethics

The research in question was not found to raise any particular ethical concerns. However, the standards practice of informed consent, confidentiality, anonymity, and privacy were ensured to be maintained all throughout the research process. Along with the poster announcing the survey, individuals were handed an information sheet delivering in depth information regarding the research project and containing the contact information of the researcher. Having provided individuals with enough information for them to be able to make an informed decision of whether or not to take part in the study, no specific consent form was devised. Instead, the first page of the PowerPoint presentation and of the survey both contained a short paragraph with information typical of a consent form (i.e. participation being voluntary, how confidentiality and anonymity were to be ensured, and the clarification of the participant's right to withdraw).

Data analysis

The results obtained from the online questionnaire were directly transferred onto a Microsoft Excel document. This provided a clear view of the data, allowing for checks to be made before the data was transferred to the software R Studio (RStudio Team, 2023), where data analysis was conducted. This process began by cleaning and preparing the data for analysis; this involved adding and removing variables, and changing the data format from wide to long, a structure deemed to be more appropriate for the planned stages of analysis. The measure D-prime (d') was used to obtain the values for recognition accuracy, the dependent variable. d' is a statistical measure developed by Wickelgren & Norman (1966) and used in experiments testing recognition memory. It is obtained by subtracting the Z-transforms of hits and false alarms, so that $d' = z(H) - z(F)$, an allows to control for sensitivity and response bias simultaneously (Keating, 2005). d' takes on values so that:

‘The highest possible d' (greatest sensitivity) is 6.93, the effective limit (using .99 and .01) 4.65, typical values are up to 2.0, and 69% correct for both different and same trials corresponds to a d' of 1.0’ (Keating, 2005).

In addition to the measure of d' , hits and false alarms were also analysed separately. Previous research has suggested doing so because of the possibility of the CRE being caused by a bias in responding that stimuli have been seen before during the facial recognition task (Slone, et al., 2000). Having obtained these scores, univariate analysis was conducted on the dependent and independent variable which was represented by the race of the stimuli. Next, bivariate, and multivariate analysis testing the relationship between the two were carried out. Techniques utilised during these stages included the comparison of groups of mean, the graphical representation of the variables, ANOVA tests, and multiple linear regression.

The following chapter will provide a detailed overview of the results obtained following the conduct of the above-described statistical analysis process on the data obtained from the recognition task.

CHAPTER 4: Results

Recognition accuracy

To avoid response bias, the dependent variable d_{prime} , indicating recognition accuracy, was obtained through the calculation of D-prime (d'). Having acquired d' scores for all participants and races of stimuli, the first step of statistical analysis consisted in the description of the dependent variable d_{prime} , graphically represented in Figure 1 below. The mean and median of this variable were extremely close to each other, namely 0.57 and 0.56, as demonstrated in Figure 1 where, because of the proximity between the two values, the red line representing the median and the blue line representing the mean, overlap. Standard deviation was 0.82, indicating low variability around the central tendency. The standard error was also low, with a value of 0.06. d_{prime} was moderately positively skewed (0.39) and had a kurtosis of 0.11, stipulating quite a flat distribution with a lack of outliers.

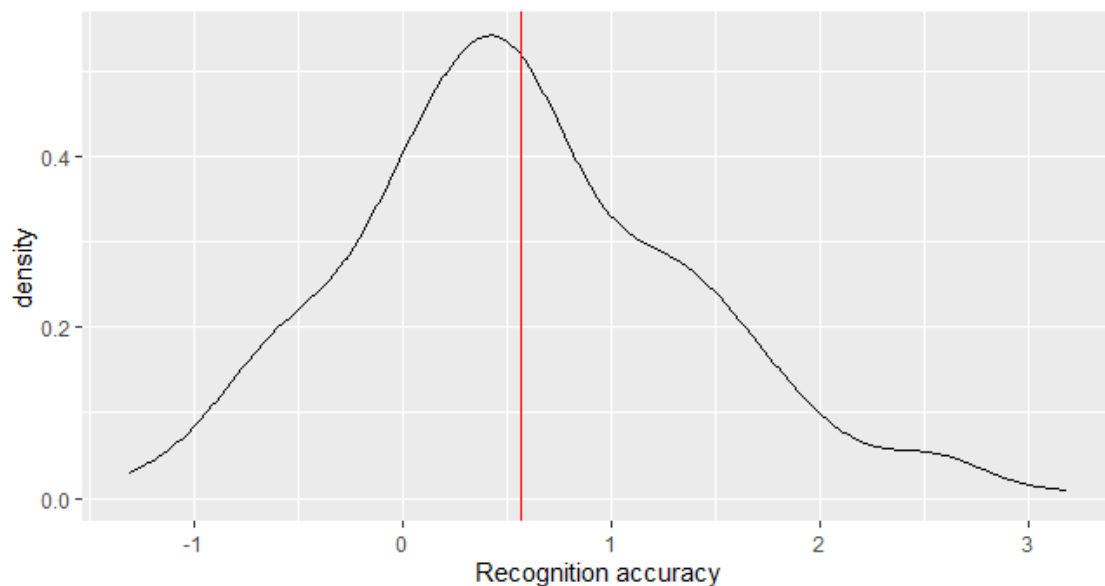


Figure 1: Distribution of variable d_{prime} indicating recognition accuracy

Hits

A description of the dependent variable $hits$ resulted in a mean of 5.35 and a median of 6.000 with low variability implied by the standard deviation which was equal to 1.92. The distribution of the variable was moderately skewed to the left (-0.83), as seen in Figure 2, and had a kurtosis value of 0.45. The standard error had a relatively low value of 0.13.

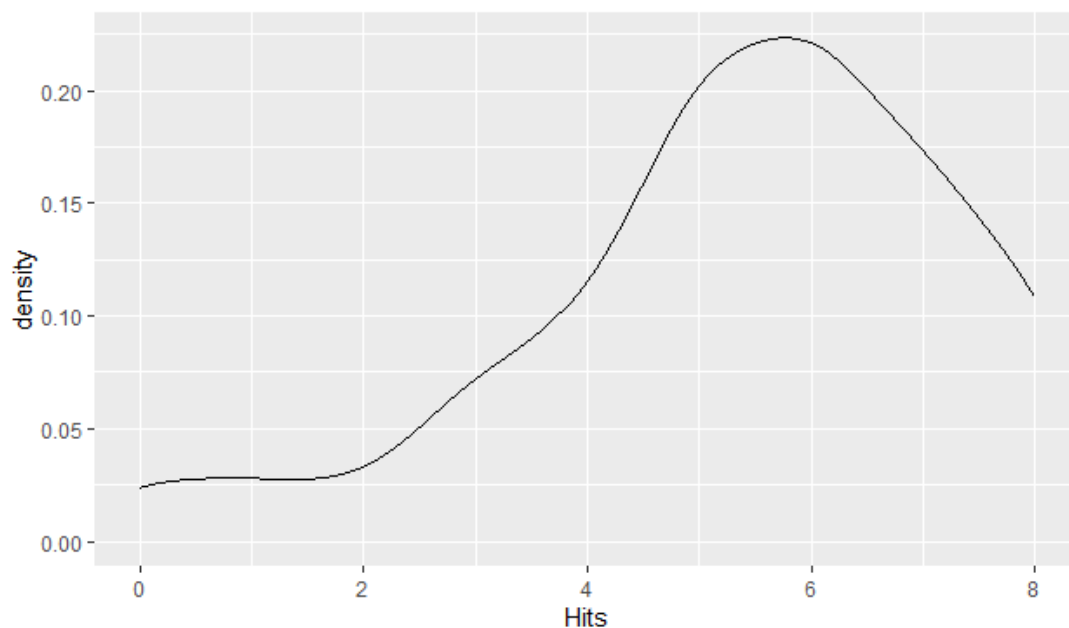


Figure 2: Distribution of hits

False alarms

The dependent variable *false_alarms* had a mean of 3.62 and a median of 3.000. As can be seen in Figure 3, the distribution was very slightly skewed to the right (0.2) and had a negative kurtosis of -0.36. The standard error was the same as it was for *hits* with a value of 0.13.

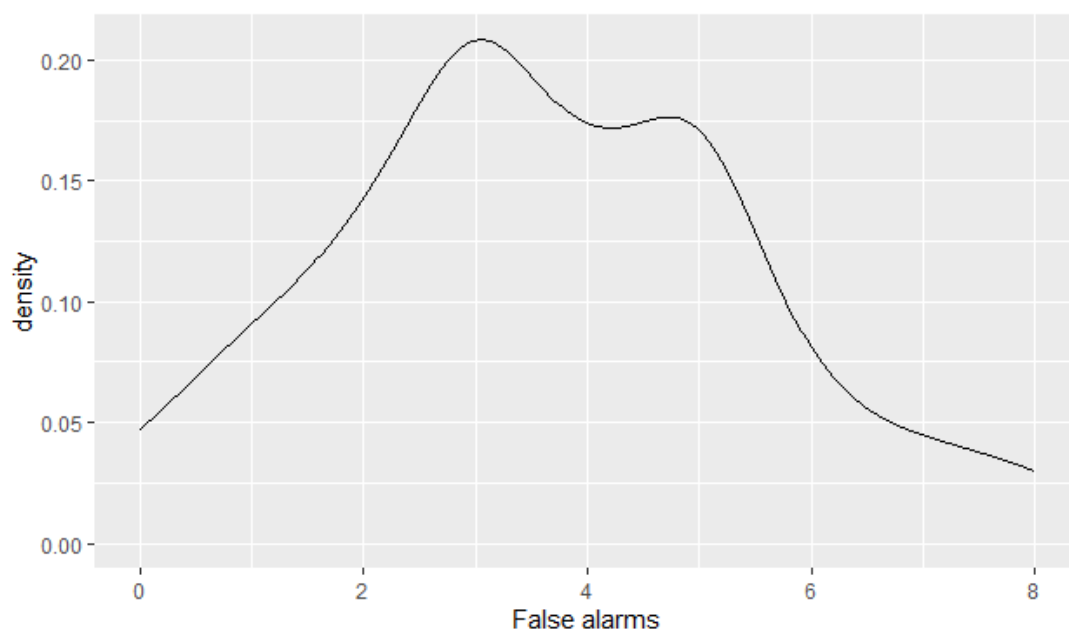


Figure 3: Distribution of false alarms

Level of contact

Contact levels for participants with the three ethnic groups considered in the study were assessed through the use of population data released by CENSUS in 2022 (Office for National Statistics, 2022).

It was assumed that the more common the ethnic group was, the higher the contact with such group would be. On the contrary, if the ethnic group was rare, this would result in a lack of contact between participants and this group.

Figure 4 represented below is a bar chart obtained from the data released by CENSUS detailing the numbers of residents belonging to 19 ethnic groups, living in England and Wales in 2011 and 2021 (Office for National Statistics, 2022). It is worth noting that the ethnic group 'White: English, Welsh, Scottish, Northern Irish or British' was removed from the chart so as to be able to identify more clearly the differences between ethnic groups which made up a smaller part of the population (Office for National Statistics, 2022).

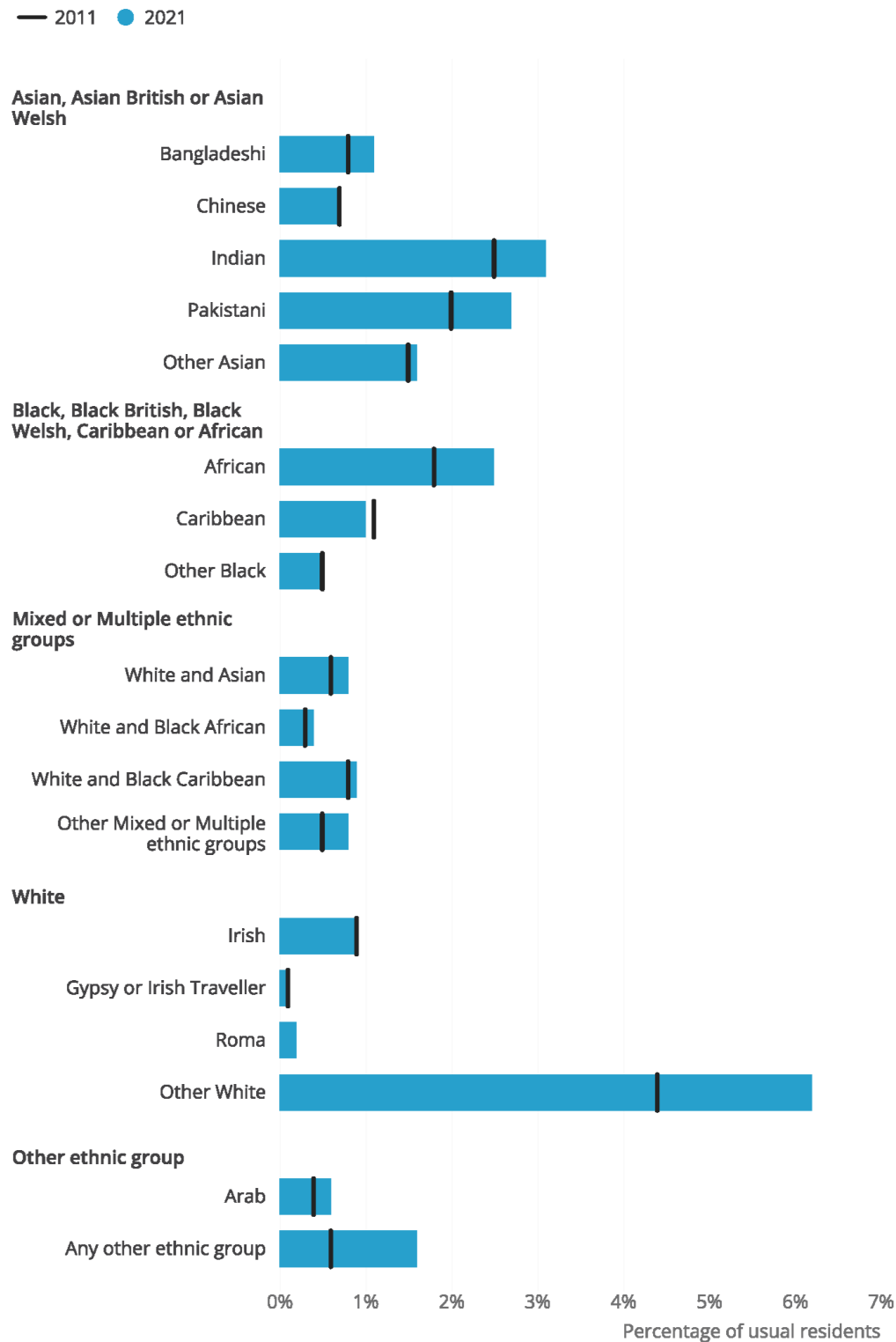


Figure 4: Ethnic minority groups, 2011 and 2021, England and Wales

(Office for National Statistics, 2022)

As shown by Table 1, the 'White: English, Welsh, Scottish, Northern Irish or British' ethnic group accounted for a total of 74.4% (44.355.038) of the population in England and Wales (Office for National Statistics, 2022). Residents who classified as 'Black, Black British, Black Welsh, Caribbean or African: Caribbean' and 'Asian, Asian British or Asian Welsh: Indian' were far less common, with Caribbean individuals in particular accounting only for 1.0% (623.119) of the total population (Office for National Statistics, 2022) (Table 1). Indian individuals on the other hand, were the most common ethnic group living in England and Wales in 2021, excluding the White ethnic group (Office for National Statistics, 2022) (Table 1).

Ethnic Group	2011 (number)	2021 (number)	2011 (percent)	2021 (percent)
Asian, Asian British or Asian Welsh: Bangladeshi	447.201	644.881	0,8	1,1
Asian, Asian British or Asian Welsh: Chinese	393.141	445.619	0,7	0,7
Asian, Asian British or Asian Welsh: Indian	1.412.958	1.864.318	2,5	3,1
Asian, Asian British or Asian Welsh: Pakistani	1.124.511	1.587.819	2,0	2,7
Asian, Asian British or Asian Welsh: Other Asian	835.720	972.783	1,5	1,6
Black, Black British, Black Welsh, Caribbean or African: African	989.628	1.488.381	1,8	2,5
Black, Black British, Black Welsh, Caribbean or African: Caribbean	594.825	623.119	1,1	1,0
Black, Black British, Black Welsh, Caribbean or African: Other Black	280.437	297.778	0,5	0,5
Mixed or Multiple ethnic groups: White and Asian	341.727	488.225	0,6	0,8
Mixed or Multiple ethnic groups: White and Black African	165.974	249.596	0,3	0,4
Mixed or Multiple ethnic groups: White and Black Caribbean	426.715	513.042	0,8	0,9
Mixed or Multiple ethnic groups: Other Mixed or Multiple ethnic groups	289.984	467.113	0,5	0,8
White: English, Welsh, Scottish, Northern Irish or British	45.134.686	44.355.038	80,5	74,4
White: Irish	531.087	507.465	0,9	0,9
White: Gypsy or Irish Traveller	57.680	67.768	0,1	0,1
White: Roma	-	100.981	-	0,2
White: Other White	2.485.942	3.667.997	4,4	6,2
Other ethnic group: Arab	230.600	331.844	0,4	0,6
Other ethnic group: Any other ethnic group	333.096	923.775	0,6	1,6

Table 1: Ethnic minority groups, 2011 and 2021, England and Wales

Based on these values, it was assumed the highest amount of contact would be with White individuals, while the lowest would be with Black Caribbean individuals. For the Indian ethnic group, it was assumed the amount of contact would lay somewhere in between that with the White and Black Caribbean groups.

Recognition accuracy across race of stimuli

The mean of 1 obtained for the category 'White' of the variable *racestim* supports the existence of a CRE in this experiment. However, the means of 0.71 for the 'Black Caribbean' category and 0 for 'Indian' category indicate a potential rejection of the alternative hypothesis speculating the validity of the contact hypothesis (Figure 5). As can be seen in Figure 5, standard deviation was highest for White stimuli, with a score of 0.94, contrary to the Indian stimuli group which had a standard deviation value of 0.54, closely followed by a value 0.59 for Black Caribbean stimuli. The trimmed means of the Black Caribbean and Indian stimuli, 0.69 and 0.01 respectively, suggested these two groups were less influenced by potential outliers compared to the White group which had a trimmed mean of 0.99. Skewness values indicated moderate skewness for all three categories. The 'Black Caribbean' category was slightly right-skewed (0.28), the 'Indian' category was slightly left-skewed (-0.22), and the 'White' category was more symmetrical (0.08) (Figure 5).

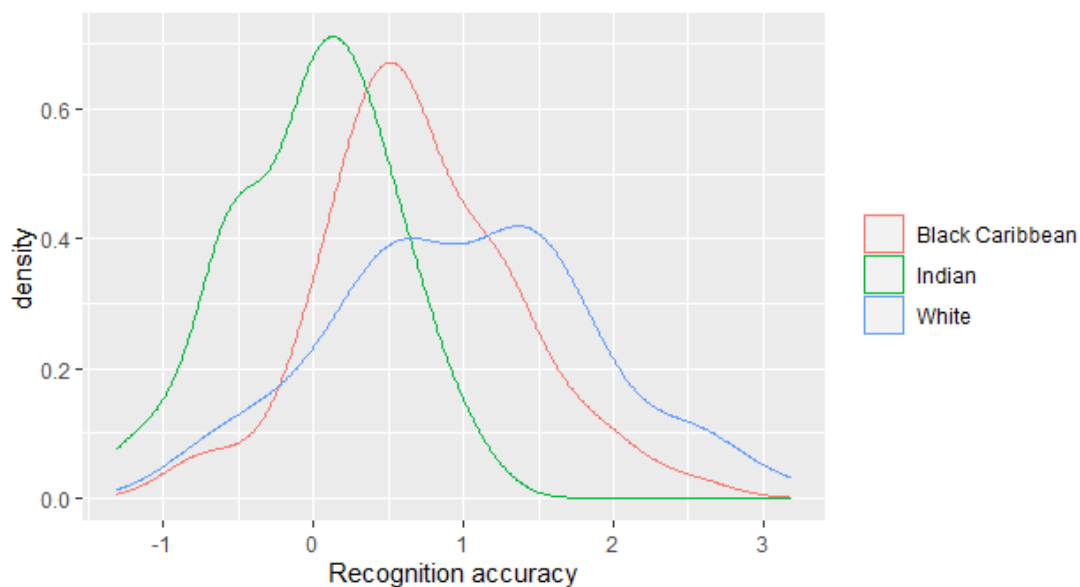


Figure 5: Recognition accuracy across the three racial groups: White, Indian, and Black Caribbean

Describing hits across races also found the category of 'White' within *racestim* to be that with the highest mean score (5.97) (Figure 6). However, this group was also that with a standard deviation of

1.91, the highest value among all three groups. The standard deviations for the other two categories were relatively close (1.74 for Indian stimuli, and 1.81 for Black Caribbean stimuli, as seen in Figure 6). The mean of the Black Caribbean group was equal to 5.59, while that for Indian stimuli had a value of 4.5. All three groups were skewed to the left, the White and Black Caribbean ones in particular, with skews of -1.25 and -1.18, respectively (Figure 6).

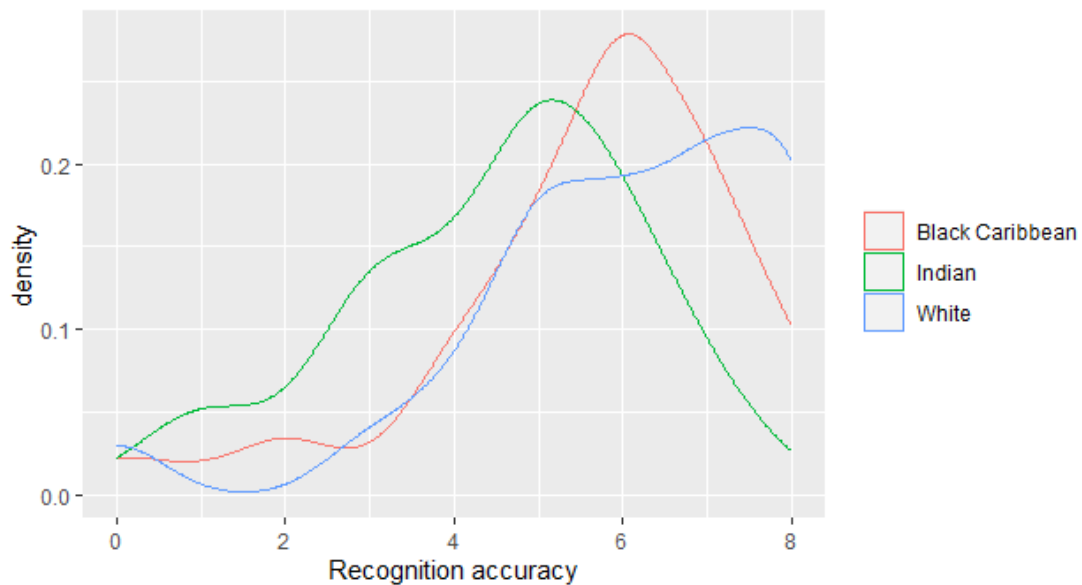


Figure 6: Hits across racial groups

The category 'Indian' of the variable *racestim* was that with the highest mean equal to 4.46. This value was followed by that of the Black Caribbean group (3.37), and finally the mean of the White group of stimuli was the lowest (3.03) (Figure 7). Standard deviation values were similar across all three groups, 1.75 for the White stimuli, 1.82 for Indian stimuli, and 1.83 for Black Caribbean stimuli (Figure 7). The White and Black Caribbean groups of stimuli were both slightly skewed to the right, with skews of 0.26 and 0.38, respectively. On the contrary, the Indian category was negatively skewed, with a skew value of -0.09 (Figure 7).

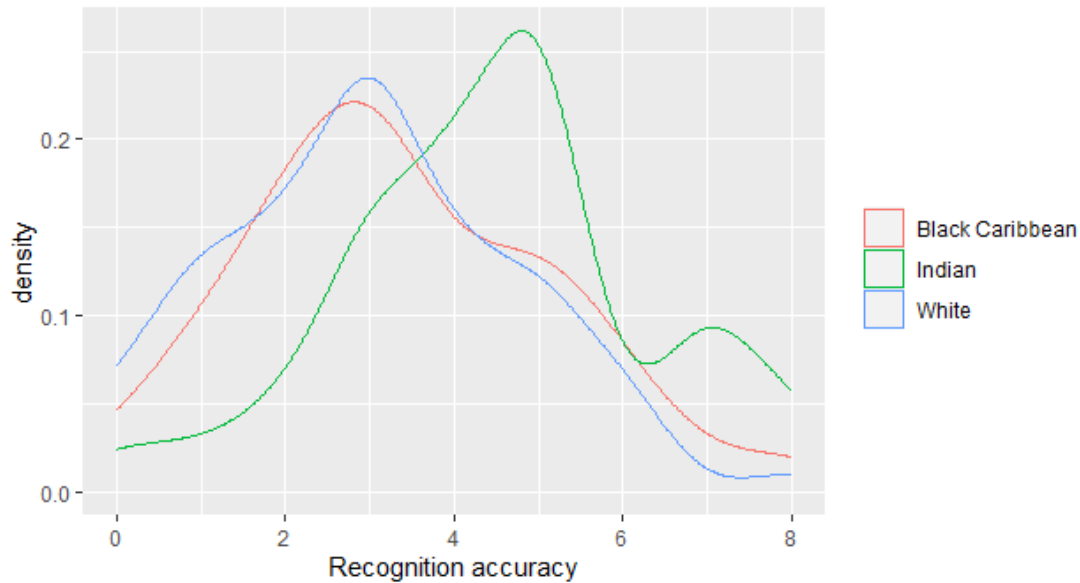


Figure 7: False alarms across racial groups

A one-way ANOVA using d_{prime} as the dependent variable produced a highly significant p-value ($p < 0.001$), confirming a significant difference of means across the three groups of stimuli. To further assess these differences, a Post-hoc Test was conducted. The highest mean difference provided by the test was 1.005, between the White and Indian stimuli groups. This value was supported by the 95% confidence interval and the significant p-value (0.000000). The mean difference between the Indian and Black Caribbean stimuli was -0.713, while that between the White and Black Caribbean stimuli was 0.291. These results were also supported by a 95% confidence interval and significant p-values, however, in the second comparison, the p-value was higher than it was for the Indian and Black Caribbean comparison (0.039636 compared to 0.000000), suggesting less of a difference between the groups.

A second one-way ANOVA replacing d_{prime} with $false_alarms$ and testing this variable across the three groups of stimuli, produced a significant p-value ($p < 0.001$). A follow up Post-hoc Test comparing the three groups of stimuli found the group of Indian stimuli to have a higher mean than both the Black Caribbean and the White group of stimuli. This result was statistically significant, 0.0013045 for the Indian-Black Caribbean comparison, and 0.0000146 for the White-Indian comparison. The 'Black Caribbean' category had a higher mean than the 'White' category. However, this result was not statistically significant, therefore it could be up to chance.

Finally, a fixed effects model testing the association between d_{prime} and $racestim$ was created. This model explained about 42.47% of the variability in d_{prime} and used 'Black Caribbean' as a reference category. The regression coefficient for the category 'Indian' was negative (-0.71) and statistically significant ($p < 0.001$) (Table 2, Figure 8), indicating a negative association between

Indian stimuli and the dependent variable d_prime . On the contrary, the coefficient for White stimuli was positive (0.29134) and statistically significant ($p < 0.01$) (Table 2, Figure 8). Therefore, in this case, the relationship between this category of stimuli and d_prime was positive. The results from this model provide support for the existence of the CRE in this experiment.

Recognition accuracy (FE)				
Predictors	Estimates	std. Error	CI	p
Indian	-0.71	0.10	-0.92 – -0.51	<0.001
White	0.29	0.10	0.09 – 0.49	0.005
Observations	210			
R ² / R ² adjusted	0.425 / 0.129			
AIC	303.333			

Table 2: Regression coefficients for model FE estimating changes for d_prime across the variable *racestim*

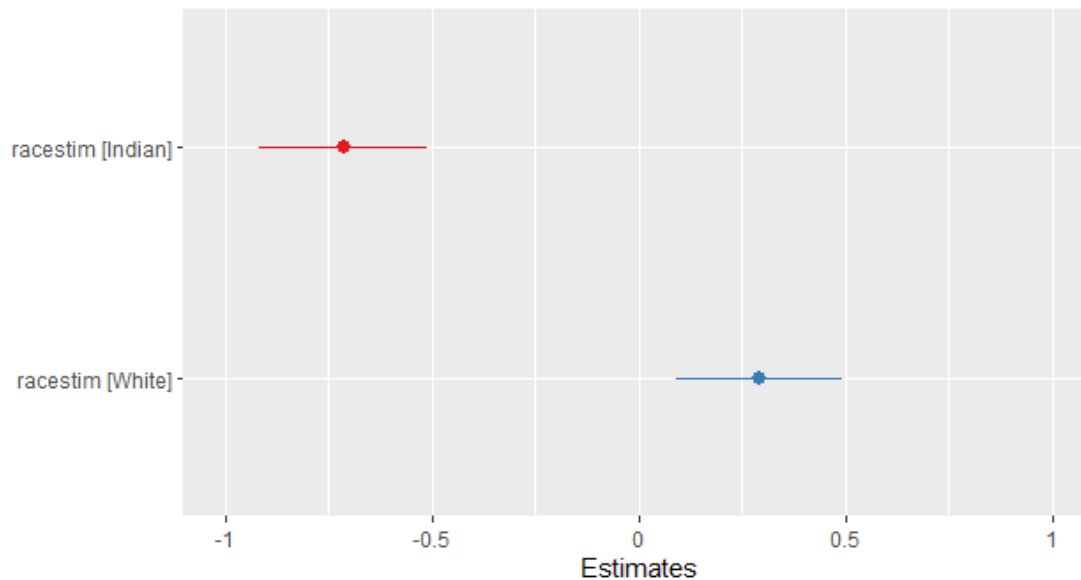


Figure 8: Regression coefficients for model FE

Recognition accuracy across age

The group of individuals who performed best was that of those aged between 45 and 54, with a mean score of 1.06, followed by the youngest group aged 18-24. Variability however was significantly high for the first group; standard deviation was 0.98 and the median absolute deviation was 0.87. This was not the case for participants aged between 18-24 years, who had a standard deviation of 0.58 and a median absolute deviation of 0.51, representing moderate variability.

The groups of participants who performed worst were those aged between 25-34 and 35-44 years, with means of 0.5 and 0.56, respectively. However, while the second group's variability was low, the first group's was high, with a standard deviation of 0.66 and a median absolute deviation of 0.46. The oldest group of individuals aged between 55 and 64 had a mean value of 0.62, a standard deviation of 0.88 and a median absolute deviation of 0.07, indicating high variability. Standard error was highest for the three older groups as they were affected by the smallest sample sizes. Kurtosis values were negative for every group, indicating that none of them were affected by extreme outliers.

To further test the association between *d_prime* and *age*, a multiple linear regression model (model 2 in Appendix A) was created, using *age* as a control variable in the relationship between *d_prime* and *racestim*. However, the output of this model proved *age* to be insignificant.

CHAPTER 5: Discussion

The findings from this research lend support to the first hypothesis speculating on the existence of a CRE among White individuals in the areas of England and Wales. However, support for the second hypothesis assuming that the CRE could be explained through the contact hypothesis even when tested on a much wider scale than usual, was not found. This chapter aims to interpret the findings of this study while acknowledging limitations which may have impacted the validity of these findings. In addition, recommendations for future research will be made.

The CRE in England and Wales

Overall, the findings from this research suggest strong differences in recognition accuracy across the three groups of stimuli: White, Black Caribbean, and Indian. It is also apparent that a CRE presented itself during the course of the experiment, a conclusion which can be made in accordance with the positive correlation between White stimuli and recognition accuracy. This group of stimuli held the highest d' (equal to 1) which, conforming to the interpretation of d' where values closer to 3 indicate a better performance and values closer to 0 indicate worst performance, suggests improved recognition for participants when identifying these faces. This finding is in line with the prediction of the CRE which implies better recognition capabilities for same race faces and is not particularly striking. Indeed, research testing the CRE has consistently demonstrated its existence (Meissner & Brigham, 2001), therefore the occurrence of the phenomenon itself does not hold as much value as the data behind its manifestation. In this research, of particular interest was the mean d' value for the Indian group of stimuli which was equal to 0. While normally in statistical analysis this would be interpreted as a uniform distribution, in this case 0 is equal to 'chance performance' (American Psychological Association, 2018) and therefore is worthy of a much more detailed interpretation. To understand this value, it can be useful to resort back to the stress Slone et al. (2000) put on the importance of measuring false alarms in studies testing the CRE. Slone et al. (2000) suggested that the CRE could be caused by a bias stemming from the likelihood of participants responding 'seen before' to faces of races different from their own. This would imply that the CRE cannot only be explained through differences in cognitive processing of faces of different races, but also through a predisposition to responding in certain ways when encountering other-race faces. In the case of this research, it could be that the occurrence of the CRE was due to this response bias on the part of participants, however this cannot be expressed with certainty, due to the limited attention given to the manifestation of bias in this experiment.

Considering the data obtained, and the way d' is affected both by hits and false alarms (Keating, 2005), it appears that in the case of recognition accuracy for the Indian stimuli, the measure was particularly affected by the likelihood of a false alarm occurring. Indeed, false alarms were the most common for participants when identifying the Indian group of stimuli, with a striking mean of 4.46. Further supporting this, is the mean value of recognition accuracy when interpreted solely in terms of hits. In fact, in this case the mean held a value of 4.5. This would suggest that the difficulty participants had was not so much in accurately identifying Indian stimuli, but more so in mistakenly identifying faces from this racial group as previously seen before. This makes it clear how the occurrence of this phenomenon is relevant in a court setting, particularly in the context of eyewitness identification (Wilson, et al., 2013). Clearly there is a predisposition for individuals to recognise faces of races different to theirs, particularly in mistakenly identifying faces as previously seen before. In a criminal justice context, this can lead to eyewitness error and potentially result in a wrongful conviction.

Measuring recognition accuracy in terms of hits only, maintained support for the CRE, demonstrating increased accuracy for White stimuli and increased inaccuracy for Black Caribbean and Indian stimuli. It is worth noting how measuring recognition accuracy without combining the approaches of measuring d' , and measuring hits and false alarms separately, adds to the findings of studies researching the CRE. Indeed, without the comparison of these two measures, it would not have been possible to gain a true understanding of the impact of false alarms on recognition.

Finally, while it was expected for age to have an impact on the relationship between recognition accuracy and race of stimuli, this was not the case. In fact, the variable was found to be insignificant for all age groups. Considering there is existing research which has successfully proven a relationship between age and recognition accuracy (Sangrigoli & De Schonen, 2004; Fioravanti-Bastos, et al., 2014), it is suggested that this result be taken with caution, as the aim of this study was not to test the significance of age in this relationship, and therefore its limited impact in this research may not mean anything.

[The contact hypothesis, an effective explanation?](#)

Contrary to what was predicted, the contact hypothesis did not seem to hold its validity in this study. Indeed, while it was expected for Indian stimuli to be better recognised than Black Caribbean faces because of their more significant presence in England and Wales, as demonstrated by the CENSUS data (Office for National Statistics, 2022), this was not the case. As mentioned, the mean for d' for the Indian category was 0 which, in the interpretation of the measure, is equal to 'chance performance' (American Psychological Association, 2018). This makes the interpretation of the d'

score for Black Caribbean stimuli more complicated. Indeed, while it could be that in the case of this group the contact hypothesis did apply, since the d' of 0.71 was lower than it was for White stimuli, there is no way of expressing this with certainty. In order to do this, it would be necessary for the d' for Indian stimuli to be comparable to that of the Black Caribbean stimuli. This comparison could be made through the sole consideration of hits, which appear to be more common for Indian stimuli than Black Caribbean stimuli, a result which would lend support to the contact hypothesis. However, this interpretation would not accurately portray recognition accuracy as, as explained by Slone, et al. (2000) and as demonstrated previously, the consideration of false alarm rates in testing recognition is extremely important.

It would not be the first time that the contact hypothesis has been found to fail. Ng & Lindsay (1994) had previously tested this hypothesis making a similar prediction to that which was made in this research. They had predicted that the minority groups involved in their study would perform better in the recognition of majority groups, compared to other individuals of their same race who lived outside of North America (Ng & Lindsay, 1994). However, while their research found evidence of the CRE, it did not find any supporting the contact hypothesis. Further research which resulted in similar findings and which involved a much more similar approach to this study was that conducted by Wong, et al. (2000), who tested the CRE in a multiracial society. In this case, the country under study was Malaysia, an area a lot smaller than the one involved in this study. Nonetheless, Wong, et al. (2000) still find that growing up in a high level multiracial society does not reduce the CRE and therefore, perhaps levels of contact do not play as much of a role as the contact hypothesis would suggest.

There are several reasons which can be taken into account when attempting to understand the reasons behind the failure of the contact hypothesis in this experiment. In their empirical paper, Wong, et al. (2020) suggested future research should consider the design of the stimuli used in the recognition task when interpreting results. Drawing from this, a first reason could be related to the previously mentioned key point made by Banks (1970) in relation to the stimuli used in recognition tasks. The author states that for distractors to be accurately compared, it is vital for these to be equivalent in all conditions (Banks, 1970). As in this research study stimuli were generated through the use of an AI tool, there is no way of ensuring this was the case. It would be assumed that having used the same prompt stimuli would be equal, however it could also be the case that the Indian stimuli generated by the AI were less diverse than the Black Caribbean stimuli. Unfortunately, due to the contemporaneity of AI tools, this can only be a suggestion, and would be a point which would be required to be further reflected upon.

A second explanation for the failure of the contact hypothesis could be associated with the factor of distinctiveness which Chiroro & Valentine (1995) and Valentine (1991), had found to be an important factor within the process of recognition. In particular, in their studies these authors were able to demonstrate how distinctive faces were more easily recognisable than typical faces. Chiroro & Valentine (1995) found that distinctiveness interacted more significantly with race when contact levels were high. In other words, high-contact individuals recognised distinctive and typical faces of all races more easily. On the contrary, low-contact individuals were more capable of recognising distinctive faces of other races compared to more typical ones. It could be the case that in the present experiment Black Caribbean stimuli were perceived as being more distinctive than Indian stimuli, and levels of contact for participants with Indian individuals were not high enough for them to have the capacity to recognise distinctive and typical Indian faces in the same way. In such circumstances, the finding by Chiroro & Valentine (1995) for low-contact individuals could have applied. That is to say, participants were better able at recognising Black Caribbean faces because they were more distinctive than Indian stimuli.

As a third point, it could be argued that the inoccurrence of the contact hypothesis during this research, and potentially others, is related to the idea of quantity and quality of contact, as had been suggested by previous research stressing the importance of these, with a particular emphasis on quality (Brigham & Malpass, 1985; Sporer, 2001; Michel, et al., 2006). While it remains complicated to assess the quality of contact, perhaps the simple condition of living in an area populated more widely by certain ethnic groups is not significant enough for the expertise developed throughout contact to develop. This could be widely due to the occurrence of self-segregation, particularly common among ethnic minorities (Daniels, et al., 2008). Segregation can be defined as:

‘The act by which a (natural or legal) person separates other persons on the basis of race, colour, language, religion, nationality or national or ethnic origin without an objective and reasonable justification’. (European Commission, n.d.)

In the case of self-segregation, it is not another person who is involved in the action, but rather the individual performing the action themselves (Daniels, et al., 2008). The segregation of these ethnic minorities can result in them experiencing less contact with other, more common, ethnic groups, leading to increased difficulty in developing expertise in recognition of other-race faces.

A final potential explanation for the failure of the contact hypothesis which is also related to the idea of quality of contact is social media influence. As has been mentioned, MacLin, et al. (2004) had suggested that contact with faces through the media could be enough to improve expertise and therefore recognition accuracy. If this were true, this could be a valid explanation for the unexpected

better recognition of Black Caribbean stimuli compared to Indian stimuli. Indeed, having more contact with these racial groups through social media would provide participants with more experience with these faces in a condition very similar to that of this experiment, having taken place online and therefore having involved the observation of stimuli through a screen.

Limitations and recommendations for future research

While this research study was able to prove the existence of the CRE for White individuals living in England and Wales, it is important to note that this type of sample provides a limitation as it restricts the generalisability of the results obtained. Studying the CRE for White individuals is particularly important as they have been found to be the group which has particular difficulty in recognising individuals of racial groups which are different to theirs (Innocence Project, n.d.). However, it is still important and worth extending this type of research to a variety of ethnic groups. It is therefore suggested that future research attempt conducting a similar study cross-culturally.

In addition, the rationale for using ethnic population statistics in England and Wales as a measure of contact between racial groups could well not be an accurate measure of contact. Living in a multiracial society may not necessarily mean that racial groups interact with one another on a high enough level to improve their recognition skills. As mentioned previously, in some cities, ethnic minorities have become segregated as they have isolated themselves from wider society, maintaining their own language and beliefs in a community created by them (Daniels, et al., 2008). It is quite clear how the occurrence of this could lead to the marginalisation of ethnic minorities, making it rarer for other ethnic groups to interact with them. Overcoming this limitation could pose more difficulty. A potential suggestion could be the additional use of a questionnaire measuring levels of contact however, as stated previously, this measure is thought to be strongly affected by bias (Walker & Hewstone, 2006). In light of this, perhaps it could be useful to conduct additional research on the population of an area, considering likelihood of ethnic segregation in addition to number of individuals of particular ethnic groups living in a certain area, before observing the CRE in the same area.

The type of stimuli employed in the study also need to be considered. The suitability of these is complicated to assess, as studies have found that different stimuli impact research in different ways. For instance, the distinctiveness of a face has already been mentioned as a potential factor affecting recognition (Valentine, 1991; Chiroro & Valentine, 1995; Meissner, et al., 2005). Other research has found differences in the way faces which include external features (e.g. hair) and faces which only include internal features (e.g. eyes or mouth) are recognised (Wong, et al., 2000). Burgund (2021) suggested the importance of the eyes of the stimuli during the process of recognition. It is believed

that AI could demonstrate itself to be particularly useful in responding to these characteristics and potentially, if given precise and extensive instructions regarding the conditions the generated faces should adhere to, create stimuli which impact the process of recognition in a much more subtle way. The popularity and continuous development of AI tools (Bengesi, et al., 2023) suggests this would be possible, however future research should first consider testing the capabilities of AI when subject to such specific instructions in the context of face recognition, a task which goes beyond the scope of this particular research.

Finally, it is suggested further research focus a lot more on bias in responding 'seen before' to other-race faces. The high likelihood of Indian stimuli being mistakenly identified as previously seen before in this experiment could be because of particular characteristics of these stimuli but could also be caused by the bias mentioned by Slone, et al. (2000). To test if this were the case, it would be useful to employ a separate measure for bias such as B²D (Donaldson, 1992) controlling for liberal (i.e. participants being more likely to respond with 'seen before') and conservative bias (i.e. participants more likely to respond with 'never seen before').

CHAPTER 6: Conclusion

The aim of this dissertation was to add to the wide body of existing research on the CRE. Because of the major implications this phenomenon has in the fields of criminal justice and criminology, and specifically in the wrongful conviction of ethnic minorities, further understanding the causes behind it is vital. Doing so could contribute to the development of policies aimed at limiting the impact of the CRE in court settings. For instance, new strategies, such as individuation training for the police, or the unbiased construction of line-ups, could be developed to be combined with the use of expert testimony in the hopes of reducing the occurrence of the CRE (Wilson, et al., 2013).

Of the many existing theories attempting to explain the causes behind the CRE, this research study chose to test the contact hypothesis. Indeed, while the contact hypothesis is known for its high reliability (Meissner & Brigham, 2001), numerous inconsistencies in research findings still exist. The choice of testing the contact hypothesis on such a wide scale (i.e. England and Wales) was made to fill the gap in research which had yet to test the validity of the hypothesis in such conditions, leading to vagueness in regards to the type of contact necessary for the contact hypothesis to work.

Overcoming the inconsistencies around the contact hypothesis could mean developing a truthful and precise understanding of one of the causes of the CRE, leading us a step closer to understanding the phenomenon. In addition, expanding the knowledge of the contact hypothesis could develop the understanding of which types of contact can improve recognition abilities across races. This could open up the possibility of 'training' individuals' recognition skills through increased interracial contact.

To test the CRE and the contact hypothesis, an online survey method was employed. A total of 70 White individuals living in England and Wales were recruited to complete a recognition task online. This included watching a PowerPoint slideshow of 24 target faces (8 White, 8 Indian, 8 Black Caribbean) and then completing an online survey asking participants to identify the 48 faces shown to them (24 target faces + 24 distractor faces, split into the same three racial groups as the target faces). The responses obtained from participants were statistically analysed in R Studio (RStudio Team, 2023) to obtain the data necessary to answer the research question. The choice of using White, Indian, and Black Caribbean stimuli in the study, and in particular the choice of sampling White individuals, was made through the information provided by the CENSUS population data (Office for National Statistics, 2022). This data revealed that White individuals were the most common in the areas of England and Wales, while Indian and Black Caribbean individuals only made up a small part of the population (Black Caribbean individuals in particular). Based on this data, it

was assumed levels of contact for participants would be especially low with Black Caribbean individuals, slightly higher for Indian individuals, and very high for White individuals.

The choice of methods for this research study were partly effective, as no contact hypothesis was observed throughout the experiment. However, it was acknowledged that this could be because of limitations born from the choice of measure of contact. This measure was found not to be extremely precise due to the non-direct relationship between predominance of ethnic group and contact with said ethnic group. To avoid this problem, it would have been necessary to include an additional measure of contact supporting the population statistics. In addition, the potential limitations of creating stimuli through the use of BingAI (Microsoft Bing, n.d.) were taken into account. These would have stemmed from the difficulties in effectively knowing how equal the stimuli were across the three ethnic groups involved in the study, making it complicated to assess whether participants recognition abilities could have been influenced by factors not considered in this study, such as distinctiveness.

Nonetheless, the research question *'To what extent can the contact hypothesis provide an explanation for the Cross-Race Effect in White individuals residing in England and Wales?'* can be answered by stating that, although there is a CRE for White individuals living in England and Wales, the contact hypothesis does not seem to provide a valid explanation for its occurrence in such conditions. Indeed, contrary to what was assumed, this study finds that Black Caribbean individuals are more easily recognised by White individuals compared to Indian individuals, the latter being more common than the former. In addition, the recognition of Indian stimuli was found to be highly affected by false alarms, suggesting a likelihood of participants to identify these faces as previously seen before. This finding is thought to potentially influence the occurrence of the CRE, making it possible to explain its verification through this response bias rather than through effective differences in cognitive processing.

These findings further confirm the validity and reliability of the CRE, but also add to the inconsistencies in research testing the contact hypothesis. In regards to the CRE, it is clear that false alarms have an extremely important role in measuring recognition accuracy and are therefore worth measuring in all research testing the CRE. It could be the case that Indian faces in particular have a higher likelihood of being misidentified as previously seen before, however further research including this ethnic group would be needed to confirm this. In particular, it is suggested future research employ a separate measure of bias to control for this possibility more precisely. In respect of the contact hypothesis, what can be drawn from these results is that quality of interracial contact does indeed appear to be more important than quantity (Sporer, 2001; Michel, et al., 2006; Brigham

& Malpass, 1985). Nevertheless, to be able to express this idea with certainty, it would be necessary to assess how effective the measure of contact used in this research study is at predicting levels of contact. Future research could aim to do this, perhaps even by considering the use of additional measures of contact.

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Appendices

Appendix A: Additional results

Model 2				
<i>Predictors</i>	<i>Estimates std. Error</i>		<i>CI</i>	<i>p</i>
Intercept	0.75	0.09	0.56 – 0.93	<0.001
Indian	-0.71	0.12	-0.95 – -0.48	<0.001
White	0.29	0.12	0.06 – 0.52	0.015
25-34	-0.15	0.14	-0.42 – 0.12	0.282
35-44	-0.05	0.19	-0.42 – 0.33	0.811
45-54	0.15	0.17	-0.18 – 0.47	0.372
55-64	-0.31	0.18	-0.66 – 0.03	0.077
Observations	210			
R ² / R ² adjusted	0.287 / 0.266			
AIC	455.179			

Appendix B: Ethics Form

Internal research ethics application form for taught student modules (where University ethical approval is in place for the module)

For modules LAW3035 covered by University of Leeds ethical approval reference [AREA 11-019]

Student ID	201538542
Your name	

Provisional title/ topic area	Attempting to apply the Contact Hypothesis to investigate the Cross-Race Effect in England and Wales.
Name of dissertation supervisor	Toby Davies

Are you planning to conduct fieldwork with (data on) human participants for your dissertation?
Enter a tick in the box next to either yes or no below.

Yes (This includes online research methods and secondary data analysis).	X
No, I am conducting library based research or content/ media analysis only.	

If you ticked 'no' you do not need to take further action in respect of ethical approval. Please proceed to the declarations on page 8 and 9.

If you ticked 'yes' you need to complete the rest of this form.

You MUST submit your signed ethics form to your supervisor upon their request.

INTERNAL RESEARCH ETHICS APPLICATION

Part A: Compliance with the module's block ethical approval

Ethical review is required for all research involving human participants, including research undertaken by students within a taught student module. Further details of the University of Leeds ethical review requirements are provided in the *Research Ethics Policy* available at:

<http://ris.leeds.ac.uk/ResearchEthicsPolicies> and at www.leeds.ac.uk/ethics.

1. Will your dissertation involve any of the following?	Yes	No
New data collected from observing individuals or populations	X	
Working with aggregated or population data		X
Any other research methodology, please specify:	Online survey	

2. Will any of the participants be from any of the following groups? (Tick as appropriate)	Yes	No
Children under 16		X
Adults with learning disabilities		X
Adults with other forms of mental incapacity or mental illness		X
Adults in emergency situations		X
Prisoners or young offenders		X
Prisoners or young offenders		X
Those who could be considered to have a particularly dependent relationship with the investigator, e.g. members of staff, students		X
Other vulnerable groups, please specify:		N/A

3. Will the project/dissertation/fieldwork involve any of the following: (You may select more than one)	Yes	No
Patients and users of the NHS (including NHS patients treated under contracts with private sector)		X
Individuals identified as potential participants because of their status as relatives or carers of patients and users of the NHS		X
The use of, or potential access to, NHS premises or facilities		X
NHS staff - recruited as potential research participants by virtue of their professional role		X
A prison or a young offender institution in England and Wales (and is health related)		X

4. Will any of the fieldwork or research take place outside the UK?	Yes	No
		X

If you have answered 'yes' to ANY of the above questions in 2, 3 or 4 then you will need to apply for full ethical review, a faculty committee level process. This can take up to 6-8 weeks, so it is important that you consult further with your supervisor for guidance with this application as soon as possible. Please now complete and sign the final page of this document. The application form for full ethical review and further information about the process are available at <http://ris.leeds.ac.uk/uolethicsapplication>.

If you answered 'no' to ALL of the questions in sections 2, 3 and 4 please continue to part B.

INTERNAL RESEARCH ETHICS APPLICATION

Part B: Ethical considerations within block ethical approval

5. Will the research touch on sensitive topics or raise other challenges?	Yes	No
Will the study require the cooperation of a gatekeeper for initial access to groups or individuals who are taking part in the study (eg students at school, members of self-help groups, residents of a nursing home)?		X
Will participants be taking part in the research without their knowledge and consent (eg covert observation of people in non-public places)?		X
Will the study involve discussion of sensitive topics (eg sexual activity, drug use)?		X
Could the study induce psychological stress or anxiety or cause harm or have negative consequences beyond the risks encountered in normal life?		X
Are there any potential conflicts of interest?		X
Does any relationship exist between the researcher(s) and the participant(s), other than that required by the activities associated with the project (e.g., fellow students, staff, etc)?	X	
Does the research involve any risks to the researchers themselves, or individuals not directly involved in the research?		X

If you have answered 'yes' to any of the questions in (5), please describe the ethical issues raised and your plans to resolve them on a separate page. Agree this with your supervisor and submit it with this form. Again you MAY be referred for light touch or full ethical review.

6. Personal safety	Yes	No
Where will any fieldwork/ interviews/ focus groups take place?		
At the university or other public place (please specify below).		X
At my home address		X
At the research subject's home address		X

Some other location (please specify below).	Online	
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If you conduct fieldwork anywhere except at the university or other public place you need to review security issues with your supervisor and have them confirmed by the Dissertation Co-ordinator who may refer you for light touch or full ethical review. A risk assessment may also be required:

<http://ris.leeds.ac.uk/HealthAndSafetyAdvice>. Write a brief statement indicating any security/personal safety issues arising for you and/or for your participants, explaining how these will be managed. Agree this with your supervisor and submit it with this form.

Please note that conducting fieldwork at the research subject's home address will require strong justification and is generally not encouraged.

7. Anonymity	Yes	No
Is there any potential for data to be traced back to individuals or organisations, for instance because it has been unanonymised or anonymised in such a way that there remains risk (eg highlighting people's positions within an organisation, which may reveal them).		X

If you have answered 'yes' to question 7, please discuss this further with your supervisor. You need to provide a strong justification for this decision on a separate sheet. This application will need to be reviewed by the dissertation co-ordinator and may require a full ethical review.

8. Data management issues

Will the research involve any of the following activities at any stage (including identification of potential research participants)?	Yes	No
a. Examination of personal records by those who would not normally have access		X
b. Sharing data with other organisations		X
c. Use of personal addresses, postcodes, faxes, e-mails or telephone numbers	X	
d. Publication of direct quotations from respondents		X
e. Publication of data that might allow identification of individuals to be identified		X

f. Use of audio/visual recording devices			X
g. Storage of personal data on any of the following:			
	FLASH memory or other portable storage devices		X
	Home or other personal computers		X
	Private company computers		X
	Laptop computers		X

If you have answered 'yes' to any of the questions under 7, you must ensure that you follow the University of Leeds Information Protection Policy: <http://www.leeds.ac.uk/informationsecurity> and the Research Data Management Policy: http://library.leeds.ac.uk/research-data-policies#activate-tab1_university_research_data_policy.

You are obliged to destroy data when your degree has been confirmed.

Dissertation Research Ethical Approval: Declaration

For students	<i>Please tick as appropriate</i>
Option 1: I will NOT conduct fieldwork with (data on) human participants for my dissertation.	
Option 2: I will conduct fieldwork with (data on) human participants for my dissertation.	X

For **options 1 and 2** - I confirm that:

- The research ethics form is accurate to the best of my knowledge.
- I have consulted the University of Leeds Research Ethics Policy available at <http://ris.leeds.ac.uk/ResearchEthicsPolicies>.
- I understand that ethical approval will only apply to the project I have outlined in this application and that I will need to re-apply, should my plans change substantially.

For **option 2** only:

- I am aware of the University of Leeds protocols for ethical research, in particular in respect to protocols on **informed consent, verbal consent, reimbursement for participants and low risk observation**. If any are applicable to me, signing this form confirms that I will carry out my work in accordance with them. <http://ris.leeds.ac.uk/PlanningResearch>)

Student's signature:

Date: 07/02/2024

For supervisors	Yes	No
<i>No further action required</i>		
I confirm that the dissertation is in line with the module's block ethical approval (Part A & question 7).	x	
I have discussed the ethical issues arising from the research with the student and agree that these have been accurately and fully addressed.	x	
<i>Further actions required</i>		
Refer to dissertation co-ordinator for further review/ discussion.		
The dissertation falls outside the module's block ethical approval and the student was advised to apply for full ethical review.		

Supervisor's signature: *Toby Davies*

Date: 8/2/2024

For office use only	Yes	No
This application satisfies the conditions for block ethical approval.		

RESEARCH PLAN

ONE copy to be handed in with your internal research ethics application and ONE copy to be given to your supervisor by [date].

Name	
Degree course	Criminal Justice and Criminology
Email address	
Research area and question: My proposed research question is 'To what extent can the contact hypothesis provide an explanation for the Cross-Race Effect in White individuals residing in England and Wales?'. While the cross-race effect has been studied for over thirty years, it still remains an area of debate as researchers attempt to give this phenomenon an explanation. Obtaining an explanation for the cross-race effect is vital in the areas of Criminal Justice and Criminology as this phenomenon has been proven to be a large cause of eyewitness misidentification and as a result, of wrongful convictions (Brigham and Malpass, 1985; Meissner and Brigham, 2001).	

Research methods:

To answer my research question, I intend to conduct an online survey coupled with the viewing of a PowerPoint presentation beforehand. During the survey, participants will be shown a number of photos of faces generated by an AI which they will then be asked to recognise further on. Once I have obtained the data from the survey, I will conduct quantitative analysis in order to answer my question.

Key sources:

Chiroro, P. & Valentine, T. 1995. An Investigation of the Contact Hypothesis of the Own-race Bias in Face Recognition. *The Quarterly Journal of Experimental Psychology*, **48**(4), pp. 879-894.

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Ng, W.-J. & Lindsay, R. 1994. Cross-race facial recognition: failure of the contact hypothesis. *Journal of Cross Cultural Psychology*, **25**(2), pp. 217-232.

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Appendix C: Participant Information Sheet

PARTICIPANT INFORMATION SHEET

ATTEMPTING TO APPLY THE CONTACT HYPOTHESIS TO INVESTIGATE THE CROSS-RACE EFFECT IN LEEDS

You are being invited to take part in this project! However, before you make this decision it is important that you read through the following information sheet that will provide you with more information about this project.

If there is anything that is not clear to you or that you would like to know more about, my contact details are provided at the end of this sheet.

Please take your time reading through this information to ensure you have understood all the details and your decision to take part or not take part in this project is fully informed.

What is the purpose of the project?

The cross-race effect refers to the idea that individuals have more difficulty recognising faces of other races compared to faces of their own race. This phenomenon has been identified as being one of the causes of eyewitness misidentification, a major cause of wrongful convictions.

The contact hypothesis is one of the many explanations that have been given in an attempt of explaining why the cross-race effect occurs. This hypothesis suggests that our ability to recognise other race faces can be influenced by the amount of interracial contact we are subject to.

With this project my aim is to understand whether or not the contact hypothesis is a reasonable explanation for the cross-race effect.

Taking part in this project will only require a short amount of your time and you will be able to participate from wherever you like as the project will take place online!

What is the reason for the listed requirements?

To successfully test the cross-race effect across different racial groups, I need 20 participants of each of the following ethnicities: Asian Pakistani, Black Caribbean, and White British.

Do I have to take part?

This is entirely up to you! If you want to take part in this project, you will need to keep this information sheet and access the first part of this project through the link provided to you at the end of this sheet.

Alternatively, if you decide to take part in this project and at a later stage change your mind, you will be free to withdraw without any constraints.

What do I have to do?

I will need you to view a PowerPoint presentation and complete an online survey. The two need to be completed in the order stated, therefore, while the presentation is directly accessible via a link provided at the end of this form, you will only be able to access the survey once you have viewed the presentation.

The presentation will show you some photos of people, while the survey will show you photos which you have previously seen in the presentation and others which you have not. During the survey, you will be asked whether or not you recognise the people. This should not take any longer than 15 minutes and after completion I will not be needing anymore of your time.

There are no requirements regarding where this survey needs to be conducted. However, make sure you are in a comfortable environment where you can focus on the project and that you complete the survey honestly. **What are the possible benefits of taking part?**

Unfortunately, I am not able to provide you with any immediate benefits. However, I hope your participation will provide me further information to add to the wide body of research regarding this topic in the hopes of minimising the occurrence of wrongful convictions.

What will happen to the data?

The data obtained from the survey will be used solely for my research project and will therefore be published along with this. The data will be safely stored on a password protected laptop and following the 24th of April will be permanently deleted.

Do I need to provide any personal information?

The only information you will be providing me with is your email address which I will be able to see when you access the presentation and the survey. I do however not need it for any specific purpose.

What will happen to the results of the research project?

The results will be presented in my research project, accessible solely to staff working at the University of Leeds. However, if you wish to access these, contact me using the details provided below!

The data obtained may even be used for the scope of future research.

Contact for further information

School of Law
University of Leeds
Leeds
LS2 9JT

Supervisor contact information:

Toby Davies
Phone Number:
Address

Thank you for your time and attention!

If you now wish to take part in this project please click on the following link to access the presentation: